

7 Canonical Transformation

The Hamiltonian mechanics does not reduce the difficulty of solving a given problem. The main advantage of Hamiltonian is that it gives a deeper insight to the structure of mechanics. It sets the stage for constructing of modern theories – statistical mechanics and quantum mechanics to name a few. There is one scenario where the use of Hamiltonian considerably reduces difficulty of the problem – cyclic or ignorable coordinates. Let say we have a system with two coordinates q_1, q_2 . Their conjugate momentum are p_1 and p_2 . So the Hamiltonian is

$$H = H(q_1, q_2, p_1, p_2).$$

If q_2 is ignorable then $p_2 = k$ is a constant. So the Hamiltonian is

$$H = H(q_1, p_1, k).$$

So the Hamiltonian depends on just two variables. Let see the same system from the Lagrangian point of view. The generalized velocities are $\dot{q}_1$ and $\dot{q}_2$. Since, q_2 is ignorable, the Lagrangian is

$$L = L(q_1, \dot{q}_1, \dot{q}_2).$$

While the Lagrangian depends on three coordinates the Hamiltonian is a function of two coordinates making it simpler to work with. It may also be possible that all the coordinates in a Hamiltonian are cyclic: in that case all the momentum are constants, $p_i = k_i$, and the Hamiltonian is imply a functions of these constants

$$H(k_1, k_2, \dots).$$

Then the Hamiltonian equation of motion for the coordinates are

$$\dot{q}_i = \frac{\partial H}{\partial k_i} = f_i(k_1, k_2, \dots),$$

which agains lead to simple solutions

$$q_i = f_i(k_1, k_2, \dots)t + c_i,$$

where c_i are arbitrary constants. While it may seem that problems where all the coordinates are cyclic are rare, there is a systematic way to convert a given problem in such form. The method is called canonical transformation. We can also appreciate the need for canonical transformation from the fact that the choice of coordinates significantly enhances the solvability of the problem. For example, to describe a particle moving in a plane, polar coordinates are significantly easier to solve the motion than cartesian. Transformation from cartesian to polar is an example of canonical transformation.

7.1 Canonical Transformation

Suppose there is a Hamiltonian described by the generalized coordinates q_i and the generalized momentum p_i : $H(q_i, p_i)$. The *canonical equations of motion* or the *Hamiltonian equations of motion* are (following (104))

$$\begin{array}{l} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{array}$$

Any set of coordinates (q_i, p_i) that satisfies the above equation is called *canonical coordinates*. Our aim is to describe the same problem using another set of coordinates (Q_i, P_i) . The only requirement to satisfy for

(Q_i, P_i) is that they must be canonical – in other words there must be a function $K(Q_i, P_i)$ such that the canonical equations are satisfied

$$\boxed{\begin{aligned}\dot{Q}_i &= \frac{\partial K}{\partial P_i} \\ \dot{P}_i &= -\frac{\partial K}{\partial Q_i}\end{aligned}}$$

The K can be identified with a Hamiltonian.

Since we are demanding that q_i, p_i, H and Q_i, P_i, K describe the same physics, the coordinates as well as the Hamiltonians must be related. The most general relations between (q_i, p_i) and (Q_i, P_i) can be written as

$$Q_i = Q_i(q_i, p_i, t), \quad P_i = P_i(q_i, p_i, t) \quad (105)$$

These are called *point transformations*.

To find a relation between the old Hamiltonian H and the new Hamiltonian K , we resort to the principle of least action. According to this, in the old description, the equation of motion follows by minimizing the action

$$\delta S = \delta \int L dt = \delta \int \left(\sum_i p_i \dot{q}_i - H(q_i, p_i, t) \right) dt = 0, \quad (106)$$

while in the new coordinates one minimizes the action

$$\delta S' = \delta \int L' dt = \delta \int \left(\sum_i P_i \dot{Q}_i - K(Q_i, P_i, t) \right) dt = 0. \quad (107)$$

The simultaneous validity of the preceding equations mean that the integrands are related by the following relation

$$\lambda \left(\sum_i p_i \dot{q}_i - H(q_i, p_i, t) \right) = \left(\sum_i P_i \dot{Q}_i - K(Q_i, P_i, t) \right) + \frac{dF}{dt} \quad (108)$$

The F is any function of the phase space coordinates with continuous second derivatives, and λ is independent if the coordinates and time. Though λ can be anything, for which $\lambda = 1$, the transformations (105) are called canonical. It is always possible to set λ to 1 by proper scaling of the coordinates.

The last term (108) contributes to the variation of the action at the end points and therefore vanishes if F is a function of some combination of old and new coordinates. It acts as a bridge between the old and the new Hamiltonian and is called the *generating function*. As we see below, once the generating function is known, the coordinate transformation rules and the new Hamiltonian can be easily found. Finding a generating function depends on the problem at hand, as well as what kind of transformation one is looking for (for example all new coordinates are cyclic etc). The guiding principles of finding a generating function are: symmetries, conserved quantities and various other physical properties of the system. There are four types of generating functions – this is related to four ways of choosing a set of coordinates and momentum. We now see what are the coordinate transformation relations for each of them.

In the first type of generating function

$$F = F_1(q_i, Q_i, t).$$

From (108) we get

$$\begin{aligned} \sum_i p_i \dot{q}_i - H(q_i, p_i, t) &= \sum_i P_i \dot{Q}_i - K(Q_i, P_i, t) + \frac{dF_1}{dt}, \\ &= \sum_i P_i \dot{Q}_i - K(Q_i, P_i, t) + \frac{\partial F_1}{\partial t} + \frac{\partial F_1}{\partial q_i} \dot{q}_i + \frac{\partial F_1}{\partial Q_i} \dot{Q}_i. \end{aligned}$$

As the old and new coordinates q_i and Q_i are separately independent the above relation holds identically if the coefficients of $\dot{q}_i$ and $\dot{Q}_i$ vanishes

$$\begin{aligned} p_i &= \frac{\partial F_1}{\partial q_i}, \\ P_i &= -\frac{\partial F_1}{\partial Q_i}, \end{aligned} \quad (109)$$

and

$$K = H + \frac{\partial F_1}{\partial t}. \quad (110)$$

If the F_1 is known the first half of the equations (109) give n relations defining p_i in terms of q_i, Q_i and t . Inverting them the n . Q_i 's can be obtained in terms of q_i, p_i and t . So the new Q_i 's are completely known in terms of old set (q_i, p_i) and t . These relations can be substituted in the second half of (109) to obtain the n P_i 's in terms of q_i, p_i and t . Finally, equation (110) gives the expressions for the new Hamiltonian K .

Example: The Hamiltonian for simple Harmonic oscillator is

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2.$$

Consider a generating function

$$F_1(q, Q, t) = \frac{1}{2}m\omega q^2 \cot Q. \quad (111)$$

Then we have

$$\begin{aligned} p_i &= \frac{\partial F_1}{\partial q_i} = m\omega q \cot Q, \\ P_i &= -\frac{\partial F_1}{\partial Q_i} = \frac{m\omega q^2}{2 \sin^2 Q}, \\ K &= H \end{aligned} \quad (112)$$

Using these relations we express the old coordinates in terms of the new coordinates. From the second equation

$$q = \sqrt{\frac{2P}{m\omega}} \sin Q.$$

Substituting in the second we get

$$p = \sqrt{2m\omega P} \cos Q.$$

Substituting in the Hamiltonian H we get the new Hamiltonian as

$$K = H = \omega P.$$

So in the new Hamiltonian the new coordinate Q is cyclic. So the momentum is P constant. As the total energy of the system E is constant, $E = K$

$$P = E/\omega.$$

From the Hamilton's EoM

$$\dot{Q} = \frac{\partial K}{\partial P} = \omega,$$

which gives a solution

$$Q = \omega t + \phi.$$

Substituting P and Q in q we get

$$q = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \phi).$$

This is the usual expression for oscillator.

The rest of the three generating functions and the transformation relations are listed in the table.

Generating Function	Generating Function Derivatives	Trivial Special Case
$F = F_1(q, Q, t)$	$p_i = \frac{\partial F_1}{\partial q_i}, \quad P_i = -\frac{\partial F_1}{\partial Q_i}$	$F_1 = q_i Q_i, \quad Q_i = p_i, \quad P_i = -q_i$
$F = F_2(q, P, t) - Q_i P_i$	$p_i = \frac{\partial F_2}{\partial q_i}, \quad Q_i = \frac{\partial F_2}{\partial P_i}$	$F_2 = q_i P_i, \quad Q_i = q_i, \quad P_i = p_i$
$F = F_3(p, Q, t) + q_i P_i$	$q_i = -\frac{\partial F_3}{\partial p_i}, \quad P_i = -\frac{\partial F_3}{\partial Q_i}$	$F_3 = p_i Q_i, \quad Q_i = -q_i, \quad P_i = -p_i$
$F = F_4(p, P, t) + q_i p_i - Q_i P_i$	$q_i = -\frac{\partial F_4}{\partial p_i}, \quad Q_i = \frac{\partial F_4}{\partial P_i}$	$F_4 = p_i P_i, \quad Q_i = p_i, \quad P_i = -q_i$

Figure 13: Four basic CTs.

7.2 Poisson Bracket

If two functions F, G depend on canonical coordinates (q_i, p_i) and time t , then the Poisson's bracket of F and G is defined as

$$[F, G]_{q,p} = \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right).$$

Many books also denote $\{ \}$ to denote PB. Also, for brevity superscripts q, p often omitted to indicate PB. From the fundamental definition, following *fundamental PB* can be proven

$$\begin{aligned} [q_j, q_k] &= 0 = [p_j, p_k], \\ [q_j, p_k] &= \delta_{jk} = -[p_j, q_k], \end{aligned}$$

where in the last line we have used the anti-symmetry property

$$[u, v] = -[v, u],$$

where u, v are any two functions of coordinates and momentum. Two other properties of PB include linearity

$$[au + bv, w] = a[u, w] + b[v, w], \quad [uv, w] = u[v, w] + [u, w]v,$$

and the Jacobi identity

$$[u, [v, w]] + [v, [w, u]] + [w, [u, v]] = 0$$

7.3 Invariance of PB under CT

In this section we will show that if $(q_i, p_i) \rightarrow (Q_i, P_i)$ is a CT then for any two function F, G

$$[F, G]_{Q,P} = [F, G]_{q,p},$$

meaning PB is invariant under CT. The proof is straightforward but lengthy. The first step is to prove fundamental PB

$$[Q_k, Q_l] = [P_k, P_l] = 0, \quad \text{and} \quad [Q_k, P_l] = \delta_{kl},$$

From the definition of PB

$$[Q_k, Q_l]_{q,p} = \sum_i \left(\frac{\partial Q_k}{\partial q_i} \frac{\partial Q_l}{\partial p_i} - \frac{\partial Q_k}{\partial p_i} \frac{\partial Q_l}{\partial q_i} \right).$$

From the generating functions F_1 and F_3 we can obtain the following relations

$$\begin{aligned} \frac{\partial p_k}{\partial Q_l} &= \frac{\partial}{\partial Q_l} \frac{\partial F_1}{\partial q_k} = \frac{\partial}{\partial q_k} \frac{\partial F_1}{\partial Q_l} = -\frac{\partial P_l}{\partial q_k}, \\ \frac{\partial q_k}{\partial Q_l} &= -\frac{\partial}{\partial Q_l} \frac{\partial F_3}{\partial p_k} = -\frac{\partial}{\partial p_k} \frac{\partial F_3}{\partial Q_l} = \frac{\partial P_l}{\partial p_k}, \end{aligned} \tag{113}$$

Substituting (113) in the PB we get

$$\begin{aligned}[Q_k, Q_l]_{q,p} &= \sum_i \left(-\frac{\partial Q_k}{\partial q_i} \frac{\partial q_i}{\partial P_l} - \frac{\partial Q_k}{\partial p_i} \frac{\partial p_i}{\partial P_l} \right) \\ &= -\frac{\partial Q_k}{\partial P_l} = 0,\end{aligned}$$

Since Q, P are independent variable. For the same reason

$$[Q_k, Q_l]_{Q,P} = \sum_i \left(\frac{\partial Q_k}{\partial Q_i} \frac{\partial Q_l}{\partial P_i} - \frac{\partial Q_k}{\partial P_i} \frac{\partial Q_l}{\partial Q_i} \right) = 0.$$

Similarly we can show that

$$[P_k, P_l]_{q,p} = [P_k, P_l]_{Q,P} = 0,$$

Finally using (113) we also get

$$\begin{aligned}[Q_k, P_l]_{q,p} &= \sum_i \left(\frac{\partial Q_k}{\partial q_i} \frac{\partial P_l}{\partial p_i} - \frac{\partial Q_k}{\partial p_i} \frac{\partial P_l}{\partial q_i} \right), \\ &= \sum_i \left(\frac{\partial Q_k}{\partial q_i} \frac{\partial q_i}{\partial Q_l} + \frac{\partial Q_k}{\partial p_i} \frac{\partial p_i}{\partial Q_l} \right) = \frac{\partial Q_k}{\partial Q_l} = \delta_{kl},\end{aligned}$$

and

$$[Q_k, P_l]_{Q,P} = \delta_{k,l},$$

from definition. The rest of the proof is given as exercise.

7.4 Equation of motion in terms of PB

The Hamiltonian equation of motion can be expressed in terms of PB. Consider a function of canonical variables and time, $f(q, p, t)$. Using the Hamilton's EoM one can write

$$\begin{aligned}\frac{df}{dt} &= \frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i + \frac{\partial f}{\partial t}, \\ &= \frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} + \frac{\partial f}{\partial t}, \\ &= [f, H] + \frac{\partial f}{\partial t}.\end{aligned}\tag{114}$$

7.5 Conditions for Canonical Transformation

Suppose we have $F = F(q_i, Q_i)$ as a generating function for a canonical transformation. Then $\partial F / \partial t = 0$ and $K = H$ and

$$p_i = \frac{\partial F_1}{\partial q_i}, \quad P_i = -\frac{\partial F_1}{\partial Q_i}.$$

Now

$$dF = \sum_i \frac{\partial F}{\partial q_i} dq_i + \sum_i \frac{\partial F}{\partial Q_i} dQ_i = \sum_i p_i dq_i - \sum_i P_i dQ_i.\tag{115}$$

This is an exact differential. Therefore, for a function F to generate a canonical transformation, it must be an exact differential. The same can be shown for other three generating functions $F = F_2(q, P) - Q_i P_i$, $F = F_3(p_i, Q_i, t) + q_i p_i$, $F = F_4(p_i, p_i, t) + q_i p_i - Q_i P_i$ - they are exact differential with respect to the coordinates q_i, Q_i .